

Problem Set 2

Due on Gradescope: check the course Drive calendar
Consult the Late Policy in the syllabus

Instructions: Work with this file as a template, perhaps by File: Download: Microsoft Word. Other options, like handwriting, are fine as long as they are legible. Name and SID at the top.

Did you work with other students? Identify them below. **If you work with others, you should write answers in your own words.** (If you understand your answers, this should be easy.)

Type your name to sign the Honor Code. Save as a PDF, upload to Gradescope, and click through to locate each question part before clicking Submit. Profit.

You will see point totals associated with each question. But we will score your problem sets based on completion, not correctness. The point totals are there to make you familiar with how the exams will work, which are scored on correctness.

Questions below:

- [1. \[18 points total\] Short questions with answers so low and a little Roman.](#)
- [2. \[40 points total\] Immigration and capital remittances in the Solow Model.](#)
- [3. \[18 points total\] The Romer Model and the Zombie Apocalypse.](#)

Did you work with other students? List them below:

No.

Please type your name as an affirmation of the Honor Code at the University of California, Berkeley.

“As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.”

Type your name:

Marcus Huang

1. [18 points total] Short questions with answers so low and a little Roman.

- (a) [2 points] Recall that real GDP measured on the expenditure side is given by

$$Y = C + I + G + NX \quad (1.1)$$

What happens if there is no government and no foreign trade? Rewrite equation (1.1) and briefly explain what it states about the disposition of income, Y . Does a component of income become interpretable as something else? Explain.

$Y = C + I$ This equation states that total income Y is divided between only consumption C and investment I . The I becomes interpretable as saving S . Then the total income $Y = C + S$

- (b) [2 points] In the U.S., what can you say about the level of the long-run share of consumption in GDP, meaning C/Y ? Briefly discuss.

The trend is the share is fairly stable over the long-run, the level is typically hovers between 68% and 70% of GDP. (2022, 2008)

From Class 2 Slides

- (c) [2 points] Consumption goods and services include things like food and health care. Investment goods and services include container cranes for seaports, architectural services, and the office buildings themselves. Do the same workers in the economy typically produce both consumption goods and investment goods? Explain.

No, consumption goods like food and health care and investment goods like office building are produced by different sectors of the economy. While some labor may be flexible in the very long-run, workers are generally specialized in their specific industries and roles

- (d) [2 points] Given your previous answers, what should be true in the short run about the U.S. share of labor engaged in producing consumption goods, if consumption and investment sectors require and benefit from the same amount of capital and technology?

If the C and I sectors use the same amount of capital and tech, the share of labor producing C goods should be proportional to consumption's share of GDP.

- (e) [2 points] Recall that when production is Cobb-Douglas with a capital share $\alpha = 1/3$, the marginal product of capital is given by:

$$MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3} \quad (1.2)$$

Suppose we knew these calibrated¹ values for A, K, and L:

A = 1,983

K = 80,823,096

L = 163

"Total Factor Productivity" is a residual

The U.S. capital stock is about \$80 trillion

U.S. employment is about 163 million

Using a calculator or a spreadsheet, insert these values into the right-hand side of equation (1.2) and report and comment on what you calculated. In words, what does your value mean? Round to the nearest thousandth = tenth of a percent and interpret.

Substitute $\bar{A} = 1983$, $K = 80823096$, $L = 163$ into $MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3} \rightarrow MPK \approx 0.105$
 This represents the gross rate of return on capital investment. Specifically, it means that adding one additional \$ of capital to the U.S. economy, given these calibrated values, would increase GDP by approximately \$ 0.105.

- (f) [2 points] Imagine that employment is reduced by 10 million workers to 153 million. Assuming nothing else changes, repeat your calculation above and report the MPK. What has happened? Why? Express the change in level and in percentage terms.

Substitute $\bar{A} = 1983$, $K = 80823096$, $L = 153$ into $MPK = \frac{1}{3} \bar{A} \left(\frac{L}{K} \right)^{2/3} \rightarrow MPK \approx 0.101$
 $\Rightarrow$ percentage change is 4%. The MPK fell because of diminishing returns, with fewer workers available to utilize the existing capital stock, each unit of capital becomes less productive at the margin. Specifically, the MPK decreased by a level of 0.004.

¹ The [Penn World Table 11.0 \(2026\)](#) reports current-price capital stock in millions of PPP adjusted 2021 US\$ (variable "cn") as \$80,823,096 in 2023, with expenditure side GDP ("cgdp") as \$25,586,132 million, for an aggregate K/Y = 3.16. In early 2026, BLS reported 163 million employed. Solow model calibration by Matthew Shapiro's comment in Eric M. Engen & R. Glenn Hubbard (2005) "Federal Government Debt and Interest Rates," [NBER Macroeconomics Annual 2004, \(19\): 83-160](#) is gratefully acknowledged.

(g) [2 points] Recall that GDP is given by:

$$Y = A K^{1/3} L^{2/3} \quad (1.3)$$

Use the calibrated values above (on the previous page) to calculate Y and Y' with A and K , and $L = 163$ and $L' = 153$. Report those levels of GDP and describe what you have found. What is the percentage change in GDP caused by this reduction in labor? Also, what was the percentage reduction in labor?

Using the calibrated values, the original level of GDP (Y) is 25586132 m, and the new level of GDP (Y') after the labor reduction is 24524424 m. This means the 10m worker reduction result in a 4.15% decrease in total GDP. In comparison, the labor force itself was reduced by 6.13%. This finding shows that output fell by less than the reduction in labor because capital and technology remained constant, allowing the remaining workers to maintain a significant portion of the total production.

(h) [2 points] Take the growth rate of both sides of equation (1.2) and briefly explain what you see.

$$g[MPK] = g[A] + \frac{2}{3}g[L] - \frac{2}{3}g[K]$$

MPK growth increases with tech and labor, but decreases with capital accumulation due to diminishing returns.

(i) [2 points] Are your answers to parts (f) and (g) and (h) consistent? Use your answer to (h) to predict what the percentage change in MPK and compare it to what you calculated. Explain.

Yes, we get $g[MPK] = g[A] + \frac{2}{3}g[L] - \frac{2}{3}g[K]$, then substitute $g[A] = 0$, $g[K] = 0$, $g[L] \approx -6.13\%$ then thus $g[MPK] \approx -4.09\%$. This matches the direct calculation from part (f) of -4% . The slight numerical difference is a standard result of growth rate rules being a linear approximation of the actual change in levels.

2. [40 points total] Immigration and capital remittances in the Solow Model.

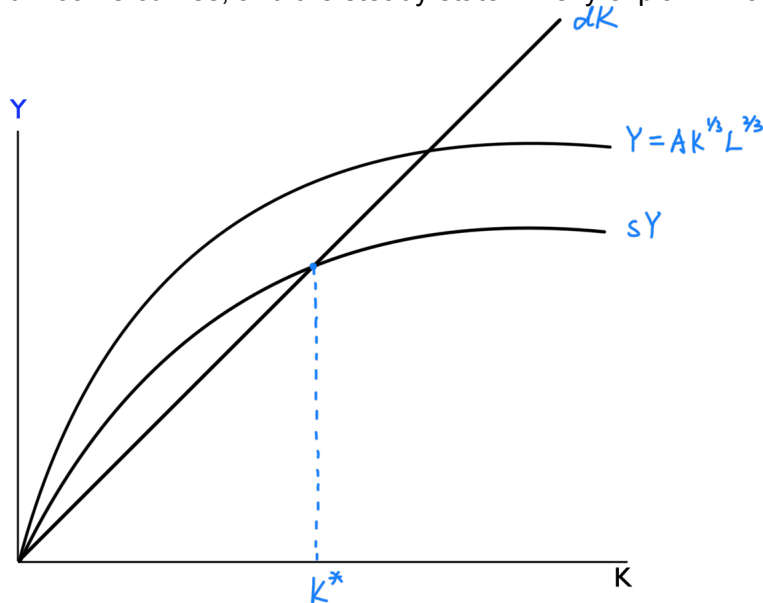
The saving rate plays a key role in the Solow Model. In particular, when the economy is closed and there are no cross-border exchanges, saving directly fuels domestic investment and increases the domestic capital stock. But in a world with mobile labor and capital, what are the implications of cross-border flows for well-being?

Imagine the U.S. is a Solow economy, and consider the effects of a one-time flow of immigration. Recall that when labor is not growing, the key equations of the Solow Model are:

$$Y = A K^{1/3} L^{2/3} \quad (2.1)$$

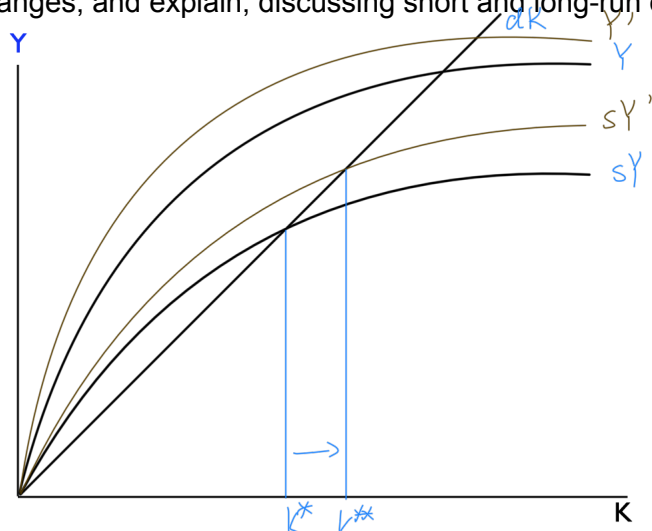
$$\Delta K = s A K^{1/3} L^{2/3} - dK \quad (2.2)$$

- (a) [4 points] On the axes below, draw the Solow diagram identifying the saving, depreciation, and income curves, and the steady state. Briefly explain what you have drawn.



- ① $Y = AK^{1/3}L^{2/3}$ showing total production
- ② sY showing the investment
- ③ dK showing capital wear-and-tear
- ④ K^* the point where $sY = dK$

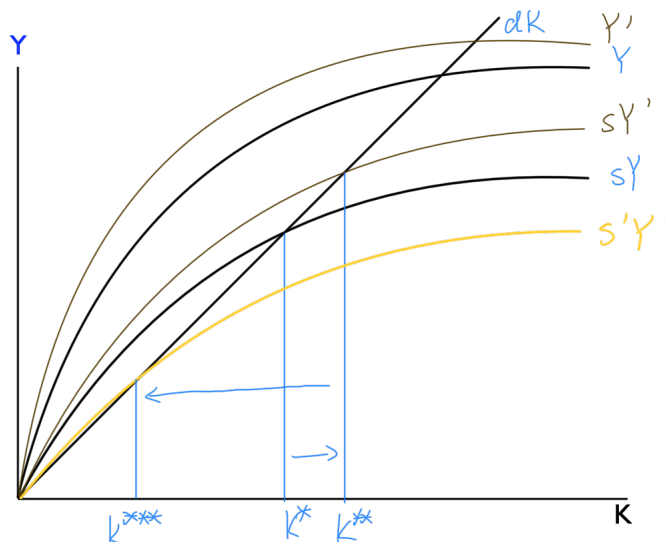
- (b) [6 points] Now redraw the Solow diagram showing the effect(s), if any, of a one-time increase in the labor supply, L , due to a substantial amount of immigration. Assume nothing else changes, and explain, discussing short and long-run effects.



- ① Shifts: Y and sY both shift upward; dK is unchanged
 ② Short Run: Total GDP rises, but output per person falls due to capital dilution
 ③ Long Run: Capital accumulates via transition dynamics to a higher steady state (K^{**}), restoring output per person to its original level.

So far, we have assumed that immigrants save at the same rate as natives, which seems reasonable. But what would happen if immigrants sent some of their saving back to their native countries, where it became foreign capital?

- (c) [4 points] Redraw part (b) showing the full effects of immigration: the increase in L and any change in the national investment rate s owing to immigration. On the next page, explain what you have drawn, discussing short and long-run effects.



- ① Shifts: Y shifts up (more L); sY shifts up (more L) but then goes down (lower s).
- ② Short Run: Total GDP rises, but output per person and wages fall immediately.
- ③ Long Run: The economy reaches a new steady state with a permanently lower level of output per person and wages because of the reduced domestic saving rate.

(d) [2 points] In the steady state, what is true about ΔK ? Rewrite equation (2.2) using this.

In the steady state, the change in the capital stock is zero ($\Delta K = 0$).

Thus: $0 = sAK^{1/3}L^{2/3} - dK \rightarrow sAK^{1/3}L^{2/3} = dK$

(e) [2 points] Substitute equation (2.1) into your answer to part (d) to find an equation that includes A , K , L , s , and d .

$$sAK^{1/3}L^{2/3} = dK$$

(f) [2 points] Manipulate your answer to part (e) to find an equation for the steady-state level of capital, K^* , in terms of A , L , s , and d . Show your work.

$$\begin{aligned}
 sAK^{1/3}L^{2/3} &= dK \\
 \Rightarrow \frac{sAK^{1/3}L^{2/3}}{K^{1/3}} &= \frac{dK}{K^{1/3}} \\
 \Rightarrow sAL^{2/3} &= dK^{(1-1/3)} \\
 \Rightarrow sAL^{2/3} &= dK^{2/3} \\
 \Rightarrow \frac{sA}{d}L^{2/3} &= K^{2/3} \\
 \Rightarrow \left(\frac{sA}{d}L^{2/3}\right)^{3/2} &= (K^{2/3})^{3/2} \\
 \Rightarrow K^* &= \left(\frac{sA}{d}\right)^{3/2}L
 \end{aligned}$$

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- (g) [2 points] Take the growth rate of both sides of your answer to part (f) to find a formula for the growth rate of K in terms of the growth rates of A , L , s , and d . Simplify.

$$\begin{aligned} K^* &= \left(\frac{sA}{d}\right)^{3/2} L \\ \Rightarrow g[K] &= g\left[\left(\frac{sA}{d}\right)^{3/2}\right] + g[L] \\ \Rightarrow g[K] &= \frac{3}{2}g\left[\frac{sA}{d}\right] + g[L] \\ \Rightarrow g\left[\frac{sA}{d}\right] &= g[s] + g[A] - g[d] \\ \Rightarrow g[K] &= \frac{3}{2}(g[s] + g[A] - g[d]) + g[L] \end{aligned}$$

- (h) [2 points] Take the growth rate of both sides of equation (1) to find a formula for the growth rate of output, Y , in terms of the growth rates of A , K , and L . Simplify.

$$\begin{aligned} Y &= A K^{1/3} L^{2/3} \\ \Rightarrow g[Y] &= g[A] + g[K^{1/3}] + g[L^{2/3}] \\ \Rightarrow g[Y] &= g[A] + \frac{1}{3}g[K] + \frac{2}{3}g[L] \end{aligned}$$

- (i) [2 points] (i) (2 points) Finally, subtract the growth rate of L from your answer to part (h) to find a formula for the growth rate of output per person, Y/L , in terms of the growth rates of A , K , and L . Simplify.

$$\begin{aligned} g[Y/L] &= g[Y] - g[L] \\ \Rightarrow g[Y/L] &= (g[A] + \frac{1}{3}g[K] + \frac{2}{3}g[L]) - g[L] \\ \Rightarrow g[Y/L] &= g[A] + \frac{1}{3}g[K] - \frac{1}{3}g[L] \\ \Rightarrow g[Y/L] &= g[A] + \frac{1}{3}(g[K] - g[L]) \end{aligned}$$

Today, the [share of Americans who are foreign-born is around 15-16 percent](#). For simplicity, let us assume that statistic is 10 percent, and although immigrants actually arrive in much smaller numbers, let us examine the effects of a one-time increase in L equal to 10 percent.

- (j) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the short run when L rises by 10 percent? Use your answer to part (i) and be specific.

It will fall, Capital is fixed, so a larger workforce dilutes the available machinery per worker.

- (k) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the long run when L rises by 10 percent? Use your answers to parts (g) and (i) and be specific.

It returns to its original level, Capital accumulation via transition dynamics eventually catches up to the labor increase.

Now let us think about potential effects of immigration on saving and capital. Assume the original saving rate s among U.S. natives was 20 percent, and let us take an extreme view: that all immigrants send all of their saving abroad. Thus their contribution to the national investment rate is zero.

- (l) [3 points] If 90 percent of residents save at 20 percent, and 10 percent of residents save at 0 percent, what is the new national or average saving rate, or is it the same? (Hint: this is like reducing the number of savers by the percent who are immigrants.)

natives: $0.9 \times 0.2 = 0.18$, immigrants: $0.1 \times 0 = 0 \rightarrow$ new national rate: $s' = 0.18 + 0 = 0.18 = 18\%$

- (m) [5 points] What are the long-run effects of immigration on output per person if the saving rate does as you have described in part (l)? Use your answers to parts (g) and (i) and be specific.

$$g[s] = (18 - 20)/20 = -0.1 = -10\% \rightarrow g[k] = \frac{3}{2}(g[s] + g[a] - g[d]) + g[l], \quad g[a] = g[d] = 0$$

$$\Rightarrow g[k] = \frac{3}{2} \times (-10\%) + 10\% = -15\% + 10\% = -5\% \rightarrow g[y] = g[a] + \frac{1}{2}(g[k] - g[l]) \rightarrow g[y] = \frac{1}{2}(-15\%) = -7.5\%$$

Long-run change: output per person falls by 7.5%

The drop in the saving rate means the economy cannot build enough capital to maintain the original capital-to-labor ratio for a large population.

3. [18 points total] The Romer Model and the Zombie Apocalypse.

Braaaaaaaains! Consider an “old school” Romer economy² like the one we saw in class, with production of knowledge A and food Y . Scientists produce new knowledge using old knowledge and have productivity $\bar{z}$. Total population is fixed at $\bar{N}$ and employed either in knowledge or in food. The share of population in knowledge production equals ℓ , a number like 0.10.

$$\Delta A_t = \bar{z} A_t L_{at} \quad (3.1)$$

$$Y_t = A_t L_{yt} \quad (3.2)$$

$$\bar{N} = L_{at} + L_{yt} \quad (3.3)$$

$$L_{at} = \ell \bar{N} \quad (3.4)$$

Recall that the solution to the Romer Model reveals the following:

$$g[A] = \bar{z} \ell \bar{N} \quad (3.5)$$

$$A_t = A_0 (1 + g[A])^t \quad (3.6)$$

$$y_t = A_0 (1 + g[A])^t (1 - \ell) \quad (3.7)$$

Imagine what might happen if all the scientists accidentally infect themselves with the “zombie flu” and become zombies. Zombies:

- Do not produce new ideas
- Do not destroy old ideas
- Eat brains and do not eat food produced by farmers
- Do not eat farmers’ brains. They prefer tasty imported brains

(a) [3 points] When the scientists become zombies, what happens to the growth rate of A ? What about the growth rate of y ?

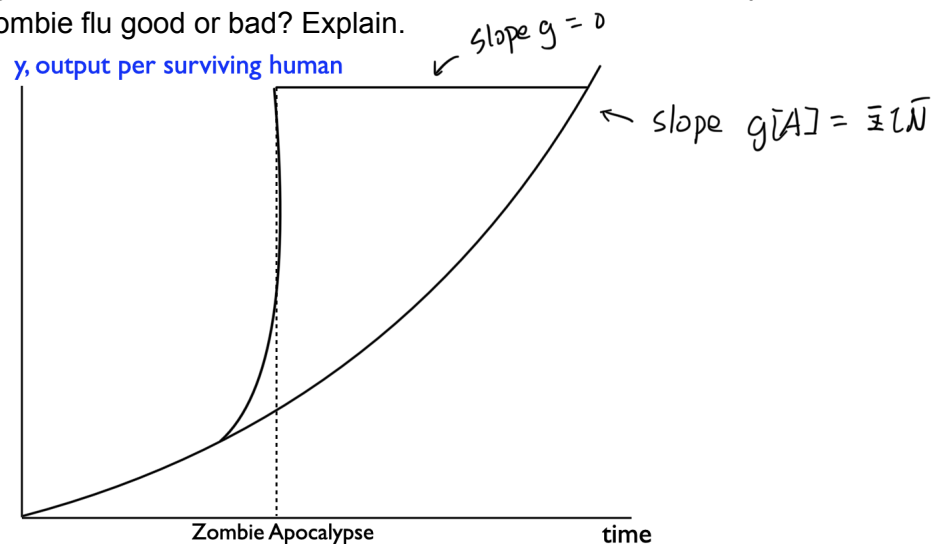
$g[A]$ falls to 0 because zombies produce no new ideas.
 $g[Y]$ falls to 0 because output growth in this model is driven entirely by idea growth.

(b) [3 points] When the scientists become zombies, what happens to the level of A ? What about the level of output per person? (Consult the bullet points about zombies above.)

① Level of A stays constant because there is no growth but no destruction of existing knowledge.
 ② Level of y stays constant because it frozen at the level reached at the time of the apocalypse.
 The “engine” of growth has stopped, but the existing “machinery” of ideas remains intact.

² The Romer model in Jones 6e is a little different. The “old school” Romer model is what appears in editions 1-5 and in the course slides for ECON 100B.

- (c) [4 points] Illustrate the Zombie Apocalypse in the graph below, showing any changes in the levels and growth rates in output per person over time. Discuss what you have drawn. Is the zombie flu good or bad? Explain.



It is bad because long-run growth is sacrificed for a one-time level increase

- (d) [4 points] Suppose rather than being turned into zombies, the scientists had all been deported to another country by the government. Would anything be different? Why?

No, the Romer model treats any loss of scientist labor as a total shutdown of the growth engine

- (e) [4 points] Suppose rather than being turned into zombies, the scientists had all been repurposed as farmers by the government. Would anything be different? Why?

No, the Romer model suggests that any shift of scientists away from idea production – even toward productive farming – destroys the long-run growth.